

Understanding :-

X.509 certificates are digital credentials that bind an entity's identity to its public key using the signature of a trusted certificate authority. These certificates form the backbone of PKI, which ensures the authentication, confidentiality and integrity in communication systems.

Firstly, I successfully implemented certificate parsing and analysis which uses X.509 certificates internal structure study to work on PEM/DER certificates parsed to extract version, serial number, key algorithms, etc. Then, I implemented certificate chain validation, which uses a trusted chain from end entity certificate $\rightarrow$ intermediate CA $\rightarrow$ root CA. Signature verification, validity period checks, and key usage validation were performed to confirm the authenticity of each link in the chain. Then, a simple certificate authority simulation was carried out to generate keys, sign CRLs, issue and revoke certificates, and maintain a certificate database. Additionally, a Certificate Transparency (CT) log demo was included to show how modern browsers detect and detect misissued certificates by auditing public logs. This experiment provides a practical understanding of how trust is been established and maintained on Internet.

Programming language r Python.